

Partial Positive Cases For Problem 2 Of arXiv:2002.12423

Automatic math-discovery smoke test

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Source Problem

The source paper is:

A. Avilés, G. Martínez-Cervantes, J. D. Rodríguez Abellán,

On projective Banach lattices of the form $C(K)$ and $FBL[E]$, arXiv:2002.12423.

Problem 2 appears in Section 4, Problems, on page 13 of the arXiv PDF.

A Banach lattice X has the property that every homomorphism into a quotient $X/\mathcal{I} \longrightarrow X/\mathcal{J}$ can be lifted to a homomorphism $\hat{T}: P \longrightarrow X$, then P is λ -projective for some λ . It is obvious that the class of ∞ -projective Banach lattices is closed under renorming but the 1^+ -projective class is not. It was asked in [10] whether every ∞ -projective Banach lattice is the renorming of a 1^+ -projective Banach lattice. But, in fact, we do not know a single example that separates these two classes, even by renorming.

Problem 1. Find an equivalent norm on a 1^+ -projective Banach lattice that makes it ∞ -projective but not 1^+ -projective.

A natural candidate would be $FBL[E]$ with E a suitable Banach space renorming of ℓ_1 .

Theorems 1.2 and 1.3 suggest a large presence of the Banach space ℓ_1 inside projective Banach lattices. This does not exclude other subspaces ($C[0,1]$ is 1^+ -projective and contains isometric copies of any separable Banach space) but we may at least ask:

Problem 2. If X is ∞ -projective and infinite-dimensional, must X contain a Banach subspace isomorphic to ℓ_1 ?

We proved that E has the Schur property if $FBL[E]$ is ∞ -projective. But the only positive case that we know is that $FBL[\ell_1(A)] = FBL(A)$ is 1^+ -projective.

Problem 3. Is there a Banach space E with the Schur property, not isometric to $\ell_1(A)$, for which $FBL[E]$ is 1^+ -projective? Is there a Banach space E with the Schur property, not isomorphic to $\ell_1(A)$, for which $FBL[E]$ is ∞ -projective?

The problem asks whether every infinite-dimensional ∞ -projective Banach lattice contains a Banach subspace isomorphic to ℓ_1 .

Result Type

This packet is a **partial result**. It proves the desired conclusion for two natural families appearing in the source paper, but not for arbitrary ∞ -projective Banach lattices.

Idea of the Proof

For free Banach lattices, the key point is that later structure theory is already far stronger than projectivity: once $\dim E \geq 2$, $\text{FBL}[E]$ contains isomorphic copies of every separable Banach space. Since ℓ_1 is separable, the conclusion follows immediately.

For $C(K)$ -spaces, no projectivity is needed either if K has a non-singleton connected component. A continuous real-valued function separates two points in that component, so its image on the component is a non-degenerate interval. Pullback along this surjection embeds $C[0, 1]$ into $C(K)$, and Banach–Mazur universality gives an ℓ_1 subspace. De Pagter and Wickstead prove that compact Hausdorff ANRs have only finitely many connected components, so the source paper’s whole $C(K)$ -projective family is covered: an infinite such K must have a non-singleton component.

Main Claim

Theorem 1. *The conclusion of Problem 2 holds in the following cases.*

1. *If E is a Banach space with $\dim E \geq 2$, then $\text{FBL}[E]$ contains a Banach subspace isomorphic to ℓ_1 .*
2. *If K is a compact Hausdorff space with a connected component containing at least two points, then $C(K)$ contains a Banach subspace isomorphic to ℓ_1 .*
3. *Consequently, if K is an infinite compact Hausdorff ANR, then the 1^+ -projective Banach lattice $C(K)$ contains a Banach subspace isomorphic to ℓ_1 . Equivalently, every infinite-dimensional 1^+ -projective Banach lattice of the form $C(K)$ satisfies the conclusion of Problem 2.*

Proof. Free Banach lattices. Remark 9.32 of Oikhberg–Taylor–Tradacete–Troitsky [2] states that $\text{FBL}[E]$ always contains a lattice copy of c_0 when $\dim E \geq 2$, and further that $\text{FBL}[\ell_1^2] \simeq C(S^1)$ contains isomorphic copies of every separable Banach space; hence so does $\text{FBL}[E]$ for every E with $\dim E \geq 2$. This supporting passage is included below.

Corollary 9.31. *Suppose E is an infinite-dimensional Banach space. Then*

$$S(E) \wedge 2 = S(\text{FBL}[E]).$$

Remark 9.32. On the other end of the spectrum, note that $\text{FBL}[E]$ always contains a lattice copy of c_0 as long as $\dim E \geq 2$, so in particular contains uniform lattice copies of ℓ_∞^n . Further, $\text{FBL}[\ell_1^2] \simeq C(S^1)$ contains isomorphic copies of every separable Banach space, hence so does $\text{FBL}[E]$ for every E with $\dim E \geq 2$. One should note, however, that $\text{FBL}[E]$ being universal is restricted to separable spaces; for other density characters it is an interesting problem to classify the subspaces of $\text{FBL}[E]$ up to isomorphism. For example, $\text{FBL}[E]$ has the same density character as E ([16, Section 3]), but, as was shown in [20], when $1 < n < 2$ and Γ is uncountable, $\text{FBL}[\ell_n(\Gamma)]$ does not embed into a

Since ℓ_1 is separable, $\text{FBL}[E]$ contains a subspace isomorphic to ℓ_1 whenever $\dim E \geq 2$. If $\dim E = 0$ then $\text{FBL}[E]$ is trivial, and if $\dim E = 1$ the free vector lattice generated by one vector has only the positive and negative parts of one generator, so its Banach-lattice completion is finite-dimensional. Thus the only infinite-dimensional free Banach lattice cases are covered by $\dim E \geq 2$.

$C(K)$ -spaces with a nontrivial component. Let K_0 be a connected component of K containing two distinct points s, t . Since compact Hausdorff spaces are normal, Urysohn’s lemma gives a continuous function $f : K \rightarrow [0, 1]$ with $f(s) = 0$ and $f(t) = 1$. The image $f(K_0)$ is

a connected compact subset of $[0, 1]$ containing both 0 and 1, hence $f(K_0) = [0, 1]$. Therefore $f : K \rightarrow [0, 1]$ is onto. The pullback map

$$J : C[0, 1] \longrightarrow C(K), \quad J(g) = g \circ f,$$

is a linear isometry. By the Banach–Mazur theorem, $C[0, 1]$ contains an isomorphic copy of every separable Banach space, in particular ℓ_1 . Thus $C(K)$ contains an isomorphic copy of ℓ_1 .

Compact Hausdorff ANR case. De Pagter and Wickstead prove that if K is an absolute neighbourhood retract in the category of compact Hausdorff spaces, then K has only finitely many connected components [3, Section 11]. If such a K is infinite and every component were a singleton, then K would be a finite union of singletons, hence finite, a contradiction. Thus an infinite compact Hausdorff ANR has a non-singleton component. The preceding paragraph applies.

Finally, the source paper proves that $C(K)$ is 1^+ -projective exactly when K is a compact Hausdorff ANR [1, Theorem 1.4]. Therefore the full projective $C(K)$ -family appearing in the source paper satisfies the conclusion of Problem 2 whenever it is infinite-dimensional. $\square$

Verification Notes

The first part is a literature-implied subcase: the supporting authors do not state that they are answering Problem 2 of [1], but their Remark 9.32 implies it for the entire $\text{FBL}[E]$ family after observing that ℓ_1 is separable and that the $\dim E \leq 1$ cases are finite-dimensional.

The second part is a standard $C(K)$ argument plus a topological theorem of de Pagter–Wickstead. The standard argument proves that any compact Hausdorff K with a non-degenerate connected component gives $C[0, 1] \hookrightarrow C(K)$. The cited finite-components theorem then ensures that every infinite compact Hausdorff ANR has such a component.

Limitations And Novelty Check

This packet does not solve Problem 2 in full. The remaining open territory is an arbitrary infinite-dimensional ∞ -projective Banach lattice that is neither visibly a free Banach lattice nor a projective $C(K)$ -space.

On June 20, 2026, web searches for the exact Problem 2 wording and close phrases such as *projective Banach lattice contains ell_1* and *infy-projective Banach lattice ell_1* located the source paper and related strong-Nakano, semiprojective, and free-lattice literature, but no explicit full answer to Problem 2. The free-lattice part is therefore recorded as an identified literature implication, not as a new proof of Remark 9.32.

Human Review Recommendation

This packet should be reviewed as a likely valid partial answer. The main points to check are the exact scope of Remark 9.32 in [2], the finite-dimensional exception for $\dim E \leq 1$, and the use of de Pagter–Wickstead’s finite-components theorem for compact Hausdorff ANRs.

References

- [1] A. Avilés, G. Martínez-Cervantes, and J. D. Rodríguez Abellán, *On projective Banach lattices of the form $C(K)$ and $\text{FBL}[E]$* , J. Math. Anal. Appl. 489 (2020), no. 1, 124129; arXiv:2002.12423.

- [2] T. Oikhberg, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices*, arXiv:2210.00614.
- [3] B. de Pagter and A. W. Wickstead, *Free and projective Banach lattices*, Proc. Roy. Soc. Edinburgh Sect. A **145** (2015), no. 1, 105–143; arXiv:1204.4282.